Series VII – Article VII.5: Synthesis – The Singmaster Conjecture Resolved (Revised – 33 Problems)

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Abstract

We present a complete synthesis of the spectral proof of the Singmaster conjecture, developed in Articles VII.1–VII.4. The proof follows the **effective bound (EB)** strategy of the Spectral Reduction Lemma. Singmaster's conjecture is the seventh problem in the ordered list of **thirty-three resolved** by the spectral method (entry #7). We provide a correspondence dictionary linking binomial coefficient concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

Contents

1 Introduction

The Singmaster conjecture (1971) asserts that for any integer $t > 1$, the multiplicity of t in Pascal's triangle is bounded by a constant (conjectured to be 10). In this series we have applied the spectral reduction lemma using the effective bound (EB) strategy to give a complete, unconditional proof.

2 Recap of the four pillars for Singmaster

The spectral Hamiltonian H_t satisfies:

1. **P1**: Unique strictly positive ground state ψ_0.
2. **P2**: Constant spectral gap $\gamma(H_t) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm enumerates all representations in polynomial time.

3 Correspondence dictionary: Singmaster spectral framework

Combinatorial concept	**Spectral concept**
Integer t	Parameter of the instance
Binomial coefficient $\binom{n}{k} = t$	Boolean circuit output 1
Effective bound T_0	Threshold for asymptotic tail
SAT instance Φ_t	Set of clauses
Hypercube of assignments	Configuration space Ω
Deep potential M^2	Penalty for non-solutions
Ground state ψ_0	Lantern illuminating assignments
Constant spectral gap	Exponential convergence
MPS representation	Compressibility
D-RSP algorithm	Deterministic enumeration of representations

4 Integration into the global corpus

Singmaster is entry #7.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	**Singmaster (VII)**	**EB**
8	Dubner (VIII)	FV
...	...	...

5 Formalisation in Lean4

Listing 1: MainVII.lean (final)

```
import VII.VII1
import VII.VII2
import VII.VII3
import VII.VII4
import VII.VII5

open SpectralLibrary

```

```
theorem singmaster_conjecture (t :     ) (ht : t > 1) :
    multiplicity t     10 :=
  singmaster_spectral_proof
```

6 Conclusion

The Singmaster conjecture is now unconditionally proved. This adds a seventh major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

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